

Plasma mediated ns-laser erosion of SiC monitored using Raman spectroscopy and in-operando LIBS

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ABSTRACT

In this study, the mechanism of nanosecond laser-assisted material erosion is revealed to be dominated by multiphoton absorption and material ejection in a gaseous plasma plume. The role of ambient gas (air or argon) in plasma removal is also reported. When cutting in air, the fact that fewer debris are found after the bigger hole forms suggest that oxygen assists material removal as gaseous products. In support of this preliminary hypothesis, the ablation mechanism was studied using a unique Raman microscope, with co-located nano-mechanical scratch testing, and with *In-situ* laser-induced breakdown spectroscopy (LIBS). From this it was confirmed that oxygen and nitrogen in air helps to gasify carbon in a plasma plume containing Si⁺, C₂ radical CO₂, CO, and CN. It was also confirmed that the structure of the ablation debris are sedimentary, with carbon network deposits overlaying Si deposits (Raman maps). The underlying SiC is in good condition with no residual stress as the ablation debris are loosely bound and can be removed by shear force. Therefore, the actual heat-affected zone is minimal. Thus, effective and clean laser cutting can be conveniently achieved with plasma-mediated erosion, in air. However, narrower cuts are best achieved under Ar.

1. Introduction

Wide bandgap semiconductor materials such as silicon carbide (SiC) and Gallium Nitride (GaN) are playing a major role in third-generation semiconductor materials for the fabrication of high voltage devices, MOSFETs and thyristors for use in extreme environments and for power amplification with high thermal resistance [1]. Due to its superior material properties such as high saturation electron drift velocity, large band gap, and critical breakdown field, SiC has gained more attention compared to silicon (Si) in the area of photovoltaics and for power electronics technology. 4H-SiC and 6H-SiC are well known amongst the more than 200 polytypes of SiC, because of their suitability for high temperature applications, which results from their α -SiC grouped hexagonal structures [2,3].

To date, several methods have been developed to fabricate micro-grooves on SiC, including micro-electric discharge, hybrid ultrasonic-machining hybrid laser-waterjet, as well as laser ablation [4–10]. Amongst these methods, those involving laser ablation are potentially

excellent options because they are contactless processes that deposits high energy density and can give high material removal rates. Being contactless, laser ablation does not cause bending stresses in the chip that are associated with mechanical processes such as sawing and thus it is more suitable for thin, hard and brittle materials. However, laser processing of SiC is difficult because of its high thermal stability resulting in a high ablation threshold.

Hybrid water-jet laser processing is attractive because the phase transition of water expansion and boiling can generate shock waves that can help to eject materials [11,12]. In addition, the jet itself can help to remove ablation products as they form and reduce the heat affected zone [HAZ]. To better comprehend these hybrid processes, it is still necessary to characterize the nature of the ablation debris and the HAZ and to have a fundamental understanding of the primary photophysical mechanisms of material ejection.

Laser ablation is achieved by focusing a pulsed laser beam onto the workpiece to deposit high amounts of energy into a very small area in a very short time. This results in extreme peak-power densities (GW/cm²/

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pulse) depending on several factors such as laser-power, pulse length, repetition-rate, scan-speed, etc. This locally deposited extreme energy vaporizes material from the workpiece via an array of mechanisms such as [13]: (i) photothermally induced phase transitions, (ii) photoacoustic shockwaves, (iii) photochemical reactions and multiphoton-induced ionization, and iv) photophysical mechanisms. It is noted that CW lasers cannot easily achieve the required peak-power densities needed for mechanisms at (ii)–(iv) [14] meaning that, for CW lasers, the material is predominantly removed via a photothermal mechanism at the same time producing large HAZs. An exception here may be CO₂ lasers, for which the wavelength of 10.6 μm may be sufficiently long for field-induced ionization to also contribute, since the Keldysh parameter (0.011 for SiC) becomes $\ll 1$ [15]. In fact, the predominance of one mechanism over the others is strongly influenced by the choice of laser, which determines the wavelength, incident power, and pulse length. But in general terms, longer and longer pulses will tend to result in single-photon photochemical events and photothermal/photoacoustic events, whereas shorter pulses produce less photothermal damage and remove material via multiphoton photophysics, generation of extreme electromagnetic fields, field ionization, etc., resulting in dramatic events such as coulomb explosion. However, longer pulses are easier to generate with high overall power densities (in $\text{mJ}/\text{cm}^2/\text{pulse}$) which can deposit more energy overall. For typical, medium-pulse energy, nanosecond processing lasers at 1064 nm a Keldysh parameter value close to 1 can be reasonably expected, meaning that, when using them, we are in neither a multiphoton ionization, nor a field-induced ionization regime, and both mechanisms are quite possible.

Although shorter pulses tend to be lower in overall power densities, they can produce much higher peak power densities (in $\text{GW}/\text{cm}^2/\text{pulse}$). Higher peak power density can easily remove material via photophysical mechanisms with no collateral photothermal damage because the overall power density is lower. Although short pulsed systems would seem to be the best choice, the cost of the technology required to generate shorter and shorter pulses increases in the order: nanosecond (Q-switched DPSS/Fiber) < picosecond (mode-lock) < femtosecond (mode-lock). Therefore, the nanosecond laser is sometimes the best choice considering cost, if we can mitigate the photothermal damage to reduce the HAZ. For a material such as SiC, which has excellent thermal stability, the concern over the size of the HAZ may be overplayed, especially if the mechanism of ablation (erosion) is initiated by multiphoton events, occurring in thermally cooler excited states via hot-plasma ejection.

Industry has been using nanosecond lasers for some time and there is generally inertia within industry regarding expensive re-fitting. By using nanosecond lasers many existing tools could be re-purposed. Additionally, even as femtoseconds and picoseconds-lasers become cheaper, so do nanosecond lasers become even cheaper. If the thermal damage typically associated with nanosecond cutting can be substantially minimized there are cost advantages that industry will appreciate. A schematic comparison of multiphoton dissociation, in excited levels, compared to thermally mediated dissociation in the ground state, is shown in SI.1. Based on the application, the size (width and depth) of laser machined grooves on SiC ranges from a few micrometers (for general applications) to a few hundred micrometers (specifically for microfluidic applications). Analytical prediction models for groove width and depth using femtosecond lasers, in air, have been established and the achievement of smooth and clean grooves in SiC with few silicon oxide particles was discussed [16]. Molian et al. [17] revealed that the mechanism of laser ablation using a picosecond laser, is coulomb explosion. In their work, pristine deep trenches were obtained and the formation of a recast layer above a laser-repetition frequency of 250 kHz was discussed. Samanth et al. [18] developed a computational model for millisecond laser machining of three-dimensional cavities in various structural ceramics considering multi-passes. Their high observed ablation rate was attributed to an increased heating rate, which was assigned as an important criterion to be considered for improved

machining efficiency. Logically, for millisecond pulses these thermal effects should dominate. Duc et al. [19] proposed a preliminary model to predict the ablation threshold of SiC using a nanosecond laser in air and found that, as expected, the ablation threshold is dependent on the absorption coefficient of SiC at the laser wavelength. Their modeling results were in full agreement with experimental results in terms of the diameter and depth of the laser drilled hole. In addition, microchannels were fabricated on 5mm-thick SiC substrates using nanosecond (100 ns, 1064 nm) laser milling for microelectronics applications, and the effect of the number of scans and predetermined width with respect to output power and scanning speed was reported [20]. A change in groove shape was observed by increasing the number of scans and no recast was detected.

Deep high-quality machining on SiC was performed by Amsellem et al. [21] using a picosecond laser with different wobble patterns, and the data was analyzed based on the effect of wobble frequency on laser power and scanning speed. Based on Raman and XRD (X-ray Diffraction) results, 2–10 μm thick amorphous re-deposits were found on the laser machined region. Wu et al. [22] introduced composite machining by coupling laser machining with plasma etching to get good surface finish of around 2.1 μm which is much less than for the simple laser machined sample. High quality surfaces were achieved as plasma etching can remove amorphous residues with low binding energy.

Nanosecond laser machining of 4H-SiC was studied by Feng et al. [23] who found that the formation of a recast layer was a critical issue. They developed a finite element model to try to understand the formation of recast humps, from which it was proposed that recoil pressure and surface tension were critical factors for material ejection and recast formation. SiC is a particularly hard material to melt, as its decomposition temperature is below its melting temperature. Indeed, it is reported that, at ambient pressure, SiC does not congruently “melt” at any temperature and either sublimates or decomposes before it melts. A high melting point of 3373 K has been calculated [24–26], but presumably Si and C would disassociate and vaporize at any temperatures above the decomposition temperature of 2800 K [27]. High pressure, which is transiently probable during inertial confinement, as a laser pulse is absorbed, may reduce the congruent melting point [25] if the inertial confinement was sufficiently long for the melting kinetics to become a possibility. However, as soon as expansion and pressure relaxation occurs, which is a prerequisite for material movement from the ablation zone, the pressure drop would enable the SiC to spontaneously decompose into gas-phase Si and C-containing radicals, which would then leave the zone as an apparent sublimation event. In the paper by Feng et al. [23], a vapor phase forming above the ablation zone is seen as critical to their mechanism. Therefore, one of the aims of the current work is to look for evidence of Si and C leaving the ablation zone in the gas phase.

While many researchers have done extensive research on the fabrication of microgrooves on 4H-SiC, there is still a lack of physics behind material removal mechanism when using nanosecond laser machining. Our present work reveals that plasma mediated material erosion is the dominant nanosecond-laser ablation mechanism, and that the carbon and silicon (respectively C₂ and Si⁺) ejected in the plasma gas redeposit sequentially as sequestered sediments in the lateral ablation debris. This can only occur with a plasma-mediated gas-phase removal mechanism. The dimensions of laser grooves created under air and argon can then be better understood, mechanistically allowing a more versatile control of the machining process.

2. Materials and methods

2.1. Wafer specifications

Based on the potential industrial applications, 4H-SiC is chosen in this study. A series of n-doped single crystal 4H-SiC (0001) wafers (produced by Gredmann Group, Taiwan) were used in this study. They

Table 1
Laser parameters used for cutting the 4H-SiC samples.

Parameter	Value
Average output power, W	8–20
Pulse energy, mJ	0.18–0.44
Power density, GW/cm ²	3.56 – 8.9
Repetition rate, kHz	45
Pulse width, ns	100
Scanning speed, mm/s	5–25
Beam quality (M ²)	< 1.5
Wavelength, nm	1064
Spot size, μm	30
Wafer thickness, μm	330
Ambience	air / argon

had a doping concentration of 10^{18} – 10^{19} cm^{-3} and an electrical resistivity of 0.015 – $0.03 \Omega \text{ cm}$. They were $50.8 \pm 0.03 \text{ mm}$ in diameter and $330 \mu\text{m}$ in thickness. The wafers were brittle, hard, optically transparent, and were light green in color. The transmittance of these wafers at the cutting laser wavelength (1064 nm) was 18.5% as shown in SI.2 as measured with a UV/Visible/NIR spectrophotometer (U-4100, Hitachi). The absorption coefficient of 4H-SiC at this wavelength was calculated as $3.99 \times 10^3 \text{ m}^{-1}$. The physical properties of 4H-SiC are listed in SI.3.

2.2. Laser system

A nanosecond ytterbium fiber laser with a wavelength of 1064 nm (IPG Photonics YLPN-1000-4 $\times$ 200-30-M) was used to cut microgrooves on 4H-SiC wafer. For this system, the Gaussian beam is triggered externally over a wide range of pulse repetition rates (2–1000 kHz) and delivers pulse energies of up to 1 mJ. The maximum peak power is 15 kW with scalable average output power up to 30 W and adjustable pulse waveforms in the range of 4–200 ns. The beam is focused on the surface of the SiC wafer with a focal length of 125 mm and a spot size of $30 \mu\text{m}$. The specific laser parameters used for cutting the 4H-SiC samples in the current experiments are listed in Table 1.

2.3. Laser cutting setup

Fig. 1 shows a schematic of the laser machining system used for cutting micro grooves on 4H-SiC wafers. The nanosecond laser from the laser source is focused onto the wafer surface using an F-theta lens after modulation. A three-dimensional worktable is designed to move the

wafer and the laser to produce a groove in the required direction. The cutting stage could be enclosed in a box with gas inlets to control the ambient gas during the machining process. In the current studies, atmospheres of air and argon were used. For experiments in argon, the gas was flowed from the bottom of the wafer, with a flow rate of 7 liters per minute. Since argon has a higher density than air, it displaces the air to produce an environment free of oxygen. Additionally, the oxygen level in the chamber was monitored using an oxygen meter.

2.4. Pre and post-processing of wafer

The wafers were cleaned ultrasonically before and after laser processing with acetone and alcohol for 15 min, and then rinsed with de-ionized water. Especially, post process cleaning was found to be effective for removal of deposits from the groove edges. Selected machined samples were encased in a resin mold and ground to see their cross sections. Subsequently, the samples were polished using 400, 800, 1200 and 2000 mesh size SiC abrasive papers.

2.5. Sample characterization

2.5.1. Optical microscopy, SEM and energy-dispersive X-ray spectroscopy

The morphology of the grooves and their cross-sections were determined using an optical microscope (OLYMPUS BX51M). Each sample was measured at three different places and the average value of groove width and depth was calculated and plotted against relevant laser parameters in subsequent analysis. In addition, the surface topography, recast formation and elemental analysis were investigated using field emission scanning electron microscopy (HITACHI SU-5000) equipped with an energy-dispersive X-ray spectrometer (American EDAX).

2.5.2. Raman spectroscopy and nano-mechanical tests

The microstructure of the materials in the laser cut grooves and the ablation debris around the grooves were characterized using a Raman microscope (Raman spectroscope, Renishaw for 532 nm). All Raman spectra were measured in a backscattering (reflectance) geometry in the range from 100 to 2000 cm^{-1} . The Raman spectra were collected with gratings of 1800 line/mm in air. The Raman spectra were recorded through an objective lens (50x, NA, 0.5, working distance 18 mm). Corresponding optical micrographs were obtained using the same microscope and set up as for Raman spectra, except using a broadband white light source to illuminate the sample in reflection mode. Raman maps are recorded over areas of $150 \times 150 \mu\text{m}^2$ (31×31 scan points). This system is unique and customized with a co-located depth sensing

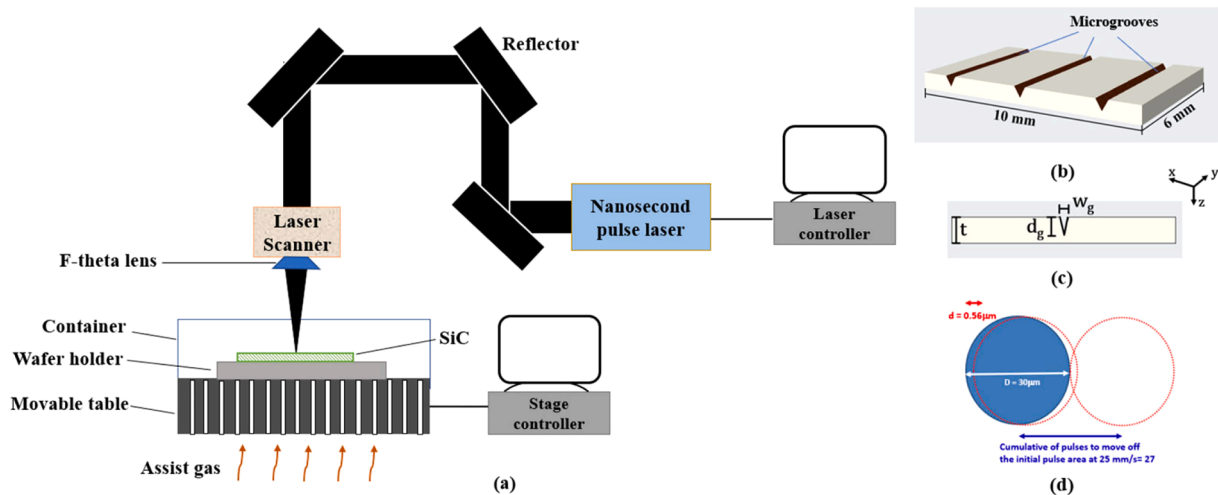


Fig. 1. (a) Schematics of the laser cutting setup (b) dimensions of wafer with grooves on it; (c) groove morphology specifications (w_g : width of groove, d_g : depth of groove, t : wafer thickness); and (d) pulse overlap.

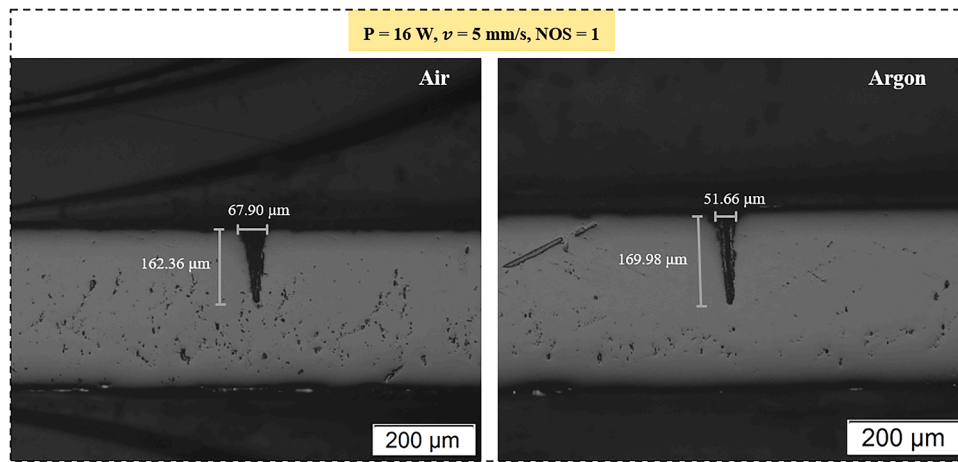


Fig. 2. Cross section of micro grooves produced in air and argon.

nano-indenter (HYSITRON TI 950 Tribo-Indenter) as described in previous works [28–32], which makes it possible to run nano-mechanical tests, or measure an SPM imaging at the exact same area where Raman spectra collected. A Berkovich diamond tip was used for depth-sensing tests. A scan rate of 0.5 Hz was used for all SPM images under a normal force of 2 μN . The nano-wear test was carried out under 583 μN normal force, with a scan rate of 2 Hz over an area of $30 \times 30 \mu\text{m}^2$.

2.5.3. X-ray diffraction

X-ray diffraction (XRD) patterns of the as-deposited and ex-operando samples were collected using a Bruker D2 PHASER X-Ray Diffractometer with a $\text{Cu-K}\alpha_{1+2} = 1.54184 \text{ \AA}$ line as a probe. The operation voltage was 30 kV, the current was 10 mA, and the collection was in the 2θ range of $5\text{--}80^\circ$.

3. Results and discussions

3.1. XRD

The high resolution XRD pattern shown in SI.4 confirms that the SiC wafers used in this study are single crystal 4H-SiC with a hexagonal wurtzite crystal structure. Two prominent peaks at $\sim 36^\circ$ and 76° are the reflections from (0004) and (0008) plane, respectively. The insert graph shows the magnified of the XRD spectrum at low intensities ($I < 450$). From this insert graph, very weak peaks at 2θ of 9, 18, 32, 45, 54.5 and 65° could be seen which are reflections from (0001), (0002), (0003), (0005), (0006) and (0007) respectively [8] and [9].

3.2. Influence of laser parameters on microgroove morphology

A detailed description of the effect of laser parameters is given in SI.5 for the power, scan rate and scan repeats, in air and in argon. However, the salient details relative to the mechanistic aspects of material removal are described below.

During laser cutting with a nanosecond laser, an ablation zone is created on the wafer surface when a high intensity laser is focused on a small area. The resulting laser ablation can be caused by photophysical, photochemical, photothermal or photoacoustic events or combination of any of these mechanisms [33]. Photochemical, photophysical and photothermal reactions can break lattice bonds, and photoacoustic processes can cause cavitation and spallation. An additional mechanism involves phase transitions causing boiling and removal of material in the gas phase. However, for a material like SiC, which is expected to decompose directly into Si and carbon radicals below the melting point, this congruent phase transition is much less likely.

If we first consider the photothermal mechanism for material removal, the temperature rise on the SiC surface after absorbing a laser pulse can be calculated using the following equation [34]:

$$\Delta T \text{ (K)} = \frac{I\alpha}{\rho C_p} \quad (1)$$

where I is the laser fluence (J/m^2), α is the absorption coefficient (m^{-1}), ρ is the density (kg/m^3) and C_p is the specific heat (J/kg/K at constant pressure – C_v can also be appropriately used depending on inertial confinement). From this, the maximum surface temperature rise caused by absorption of a single laser pulse in a linear process is estimated to be $\sim 570 \text{ K}$ for an average laser power of 10 W. In this case it would take the absorption of at least 5 pulses (with no cooling time in between each) in order to reach the vaporization temperature of SiC. Therefore, we can reasonably expect that it would require heating by >5 overlapped pulses, to cause pressure and temperature jumps close enough to the decomposition condition, to induce thermally induced ablation. In fact, our sequential laser pulses are highly overlapped, and within the scan rates used more than 30 pulses could arrive before the scan moved completely off the area of the first pulse (See schematic in SI.6). However, cooling would occur and from literature on similar dimension systems with similar repetition rates we can conclude that the maximum cumulative ΔT is unlikely to exceed 2 to 3 times the initial single pulse ΔT [35–37].

In Fig. 1(c) we can see that for a sequence of 5 consecutive pulses, the overlap significantly deviates from the first to last pulse. This means it is unlikely that a purely thermal erosion mechanism can be dominating. This means that photophysical and multiphoton processes must also be considered as distinct possibilities. These processes can result ionization, which produces plasma, which expands due to Coulomb repulsion transferring material away from the ablated zone creating an ablated groove in a plasma gas. If plasma mediated gas phase ejection is a dominant factor, then we can reasonably expect that the choice of ambient gas should change the results.

Looking at the optical microscopy images in SI.7, compared with ablation in argon, ablation in air created grooves with more volume, which can be due to gasification of carbon into CO_2 . In argon, a darker apparent HAZ was seen, while in air, a less dark wider apparent HAZ was identified. The darkness should relate to thickness and the carbon content of the debris, and the width should relate to mobility of the ablation debris as they eject. But in summary, from these results, we may expect that a thicker and less mobile debris field, which is richer in carbon, forms for the ablation under argon. This will be later confirmed from Raman mapping analysis (*vide infra*). The apparent HAZ is defined horizontally from the border of the groove to the end of the surface modified (dark-colored) region. This apparent HAZ is termed “apparent”

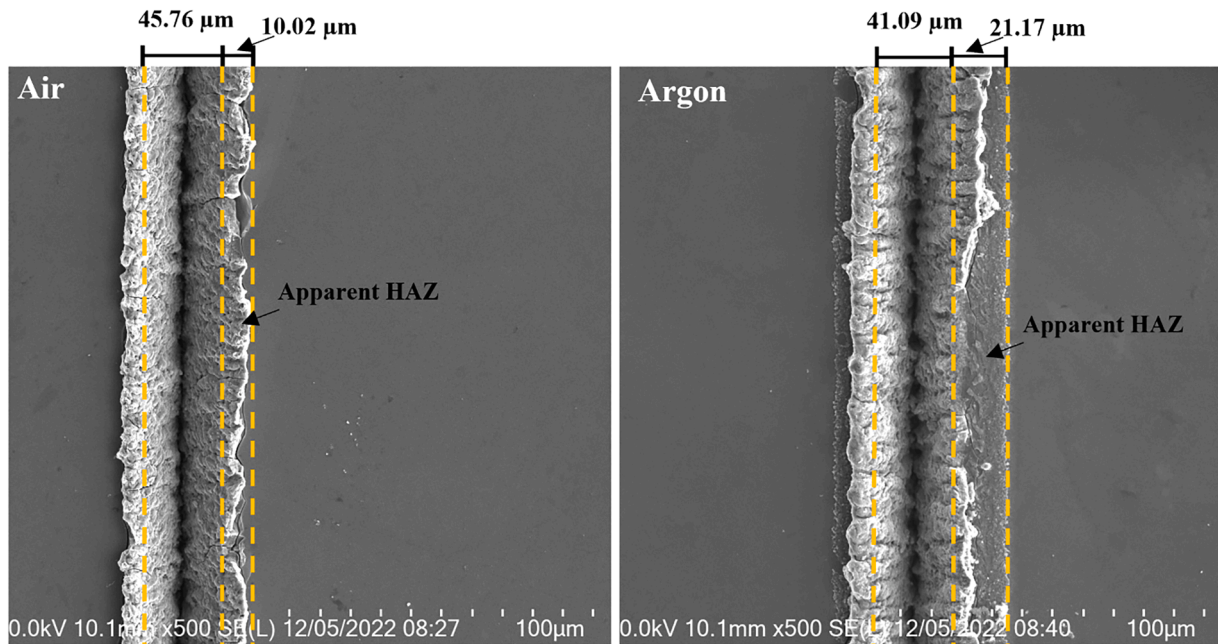


Fig. 3. SEM images of SiC grooves in air and argon.

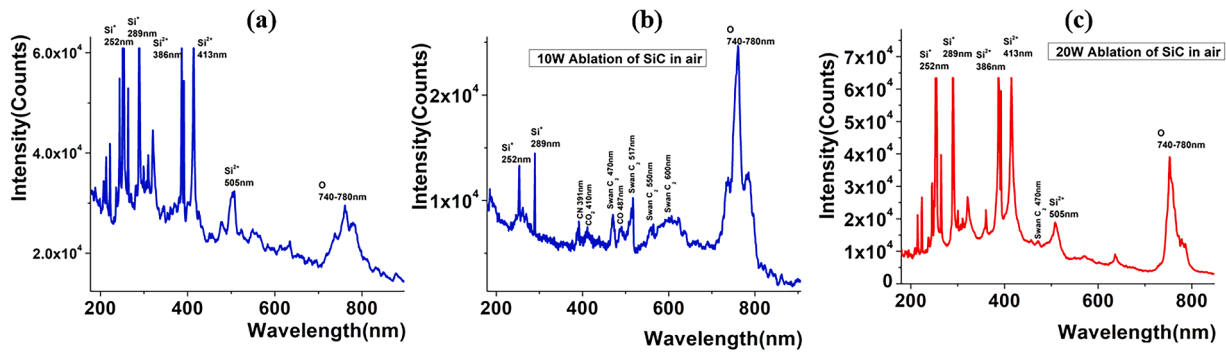


Fig. 4. LIBS spectra collected during laser processing of (a) Si at 10 W in air, and for SiC in air, (b) at 10 W, and (c) at 20 W.

because it is mainly formed by ablation debris, rather than thermally modified SiC, as will be described later.

Since the groove edges are sequentially (from top-down) coated in carbon, and Si (*vide infra*) the groove width determined from the top view is imprecise. Therefore, cross sections were used to calculate the precise depth and width of grooves, as illustrated in Fig. 2. From this, the groove depth in argon was found to be only $\sim 3\%$ more than that it is in air, which may simply be due to some small differences in optical opaqueness of the plasma plume during ablation. Besides, the groove width in air is 30 % wider than that it is in argon, meaning that there was a greater volume of material removed in air compared to in argon. Factors such as average output power, scanning speed, and number of scans play crucial roles in the formation of micro grooves.

3.3. SEM analysis

Fig. 3 shows SEM analysis of laser grooves fabricated on a SiC wafer at 10 W in both air and Ar ambiances. The actual structure of the apparent HAZ is more detailed than is evidenced from lower magnification optical images in SI.7. The Dark central bands seen in SI.7 correspond to the marked apparent HAZ in the SEM images. And as for SI.7 this is wider for argon samples. The darker diffuse extended apparent HAZ seen in SI.7 is not obvious in the SEM images, so the sensitivity of the optical image is greater for that diffuse material, maybe

indicating it is a highly absorbing black material such as carbon. Moreover, the recast deposit contains relatively less Si and more carbon in argon and more Si and less carbon in air. These results are in good agreement with Raman maps as well as LIBS data, because we show that CO_2 and CO form in air.

3.4. In-situ LIBS of laser-produced plasma

Laser-induced breakdown spectroscopy (LIBS) [38] is helpful for identifying gas-phase ablation products by analyzing the emission from laser-induced plasma, and hence it is excellent for determining the basic mechanism of laser ablation. In this work, LIBS is used to characterize the plasma produced during laser ablation of SiC wafer in air, and the results of this are shown in Fig. 4. Fig. 4(a) shows the plasma spectrum produced during laser ablation of Si in air at 10 W. This locates the positions of the expected Si^+ peaks that we may find in SiC breakdown, which are labelled in Fig. 4 [39]. The position of the expected peaks associated with carbon C_2 radicals are well known as the Swan bands and have been well reported in [39–42].

Fig. 4(b) and 4(c) respectively show the 10 W and 20 W LIBS breakdown spectra of SiC in air. Fig. 4(b) shows that SiC breaks down to form an abundance of C_2 radicals, but also Si^+ radicals. This is expected as SiC would logically split upon ablation in order to gasify and leave the ablation groove and go into a plasma plume. In addition, we also see the

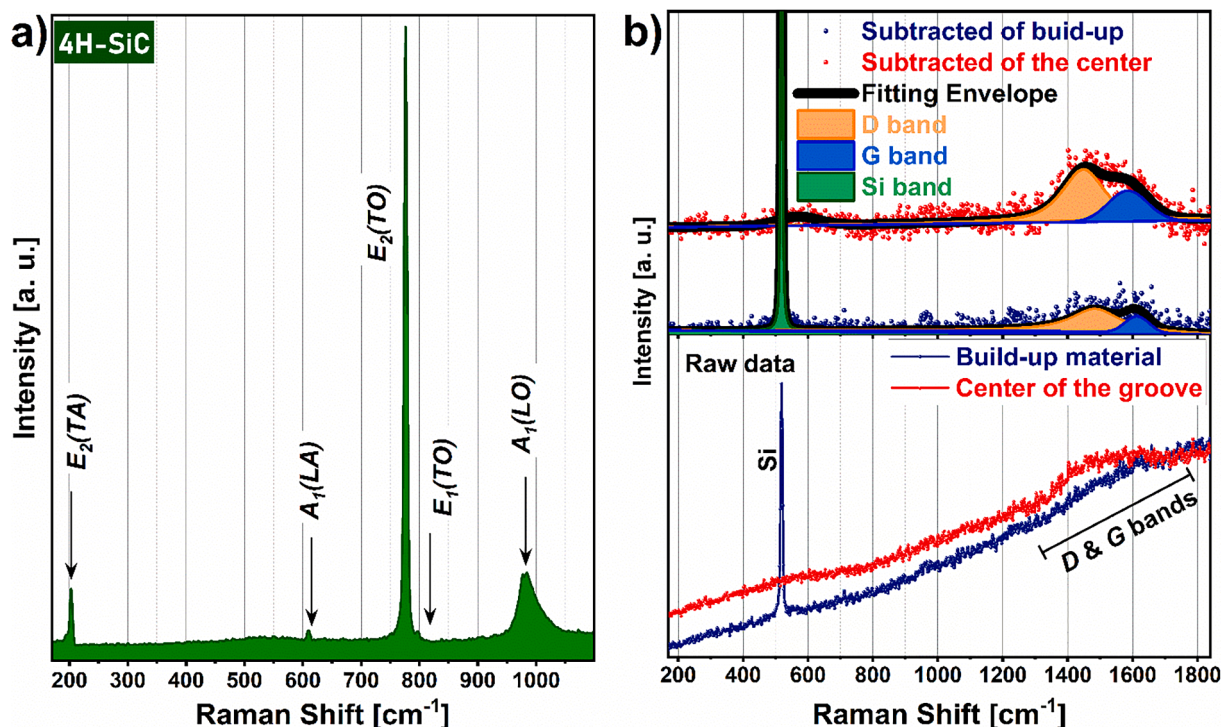


Fig. 5. (a) The Raman spectrum of 4H-SiC wafers and (b) The as-collected and PL background subtracted Raman spectra over ablation debris in the vicinity of a laser cut groove (laser power of 10 W in air).

presence of CN, CO and CO₂ which strongly indicates that oxygen and nitrogen in the air may assist in the gasification process, which we further expect will improve the material removal rate. Fig. 4(c) shows that more Si⁺ radical is seen at higher power. This is no doubt because the vaporization temperature of Si is higher than the temperature required to gasify carbon [27] and more power is required to move it into the plasma plume. Hence, we see relatively more Si⁺ at 20 W and relatively less C₂.

3.5. Determination of plasma temperature

The plasma temperature was calculated from the relative intensities of C₂ and Si⁺ vibrational bands-based Boltzmann's distribution [43]. Before the temperature was calculated, the intensities were corrected for the detector response efficiency (SI.8) with wavelength and transition probability tables from NIST (SI.9). The calculated temperatures represent the average temperature that existed during the lifetime of that species in plasma. Species present in the early plasma should experience higher average temperatures than species that exist in the plasma for longer as it cools. This gives a rudimentary assessment of the relative residency of the species in the plasma plume.

Not surprisingly, the average temperature from C₂ peaks was estimated as ~3500 K, whereas the temperature from the Si⁺ peaks experienced almost ten times higher. This indicates that the Si⁺ lifetime in the plasma is shorter and it exists only in the early phase of the plasma plume when the plasma temperature is high (See Fig. 8). On the other hand, C₂ existed in the plasma for longer, where it cools. We can therefore expect that, if the Si⁺ and C₂ leaves the plume via condensation then it should deposit as lateral debris sequentially, with Si depositing underneath the layers of carbon that arrive later. This can be tested using Raman spectroscopy of the lateral debris, as we describe later.

Another salient point from the temperature assessment of the plasma is that it way exceeds the possible temperatures that could be produced from substrate heating based on Eq. (1). This means that the origin of the

hot plasma cannot be energy transfer from a hot substrate because temperature cannot migrate in an uphill gradient. The origin of the plasma temperature can therefore only be explained by multiphoton ionization (as in SI.1) which directly migrates superheated plasma gas from the substrate material, which is relatively much cooler.

3.6. Raman spectroscopy

The 4H-SiC polytype has a hexagonal wurtzite crystal structure. Fig. 5(a) shows Raman spectra of the single crystal 4H-SiC wafers collected in the range of 140–1120 cm⁻¹. The bands at ~204, 612, 777, 798 cm⁻¹ can be assigned to E₂ planar or transverse acoustic (TA), A₁ axial or longitudinal acoustic (LA), E₂ transverse optical (TO), and E₁ transverse optical (TO) modes [44,45].

Fig. 5(b) shows as-collected and PL background subtracted Raman spectra over ablation debris in the vicinity of a laser cut groove (at edge of the groove) and the material in the center of the groove, cut in air with laser power of 10 W. Both spectra are dominated by a strong photoluminescence (PL) background. Since signals from some bands may be visually swamped by the strong PL background, it has been subtracted and the resulting subtracted spectra are presented. The presence of the Si band is obvious in the spectrum of the ablation debris. This band is absent in the spectrum of the center of the groove. The absence of this band in the center of the groove hints that the nature of the ablation debris, which differs from the material inside the grooves. After PL background subtraction, the signals from a-C network, with typical D and G bands, become obvious. The strong PL background is reported in the Raman spectra of Si-doped amorphous carbon films (a-C) and hydrogenated amorphous carbon films (a-C:H) and is believed to originate from radiative recombination of photo-excited electron-holes in isolated sp² domains in sp³ carbon networks [31,46,47,48]. Isolation of sp² domains increases the probability of radiative electron-hole pair recombination whilst reducing the chance that the hole or electron can migrate to an impurity site where non-radiative decay could occur.

A series of Raman maps were made on 4H-SiC wafers over the laser

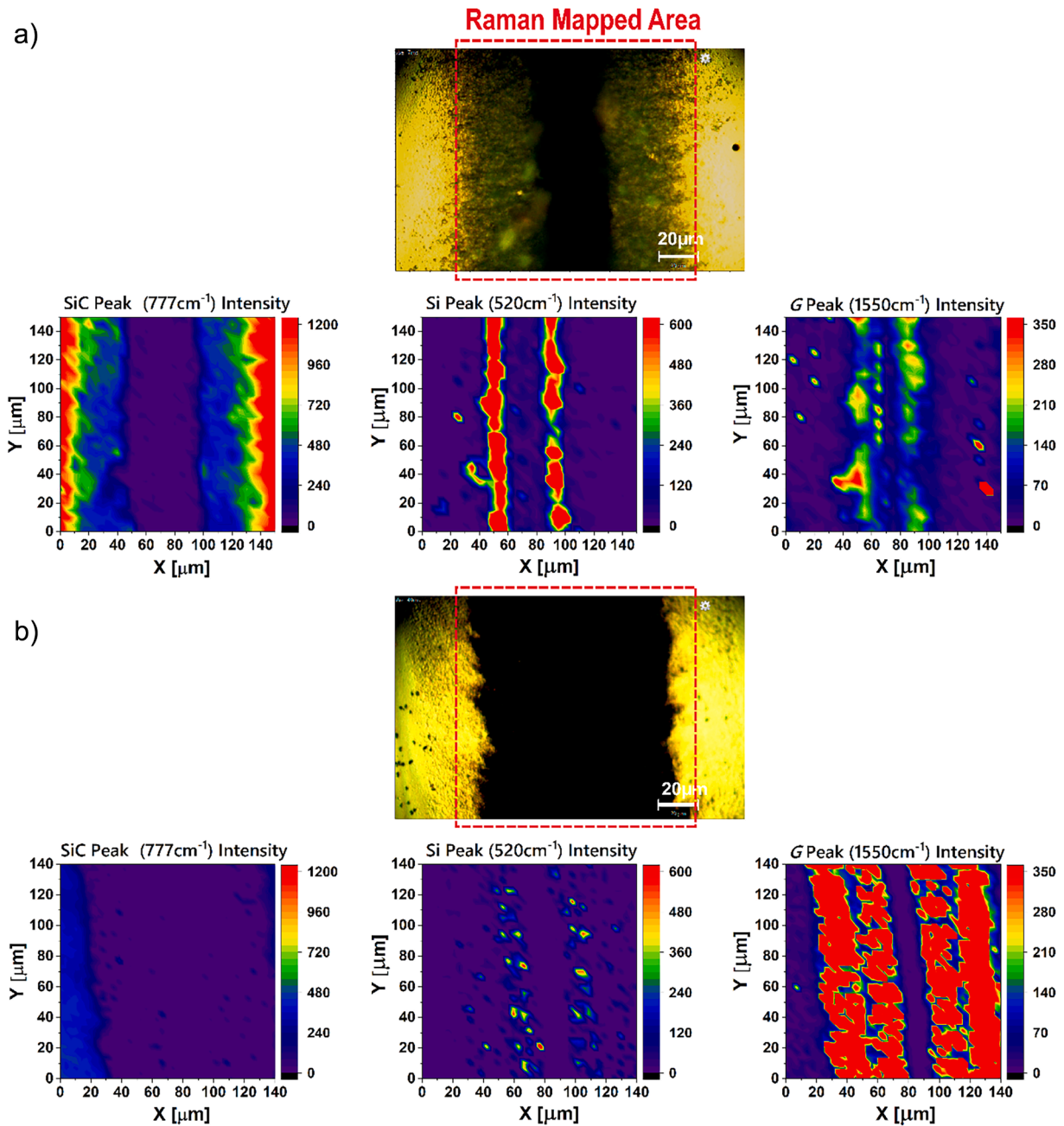


Fig. 6. Optical images over the grooves cut with 10 W laser power in (a) air and (b) in argon, illustrating the Raman mapped area along with corresponding Raman maps based on E_2 (TO) of SiC, G and Si bands intensity.

cut grooves, which were cut in different atmospheres of air and Ar, to get more insights on the mechanism of events happening during laser ablation. Fig. 6(a) and (b) shows optical images over the laser cut grooves with a laser power of 10 W in air and Ar, respectively, along with corresponding Raman maps based on E_2 (TO) of SiC, G and Si bands intensity over areas of $150 \times 150 \mu\text{m}^2$ (31×31 scan points).

From the Raman maps, we can see that there is less Si detected in the ablation debris for the sample cut in Ar and more carbon is detected in the same region, which strongly suggests that the Si layer is underneath the carbon layer. This tends to confirm the conclusions that we drew from LIBS. LIBS showed that the main components of the ejected material in the plasma are Si^+ radicals, C_2 radicals and products of reactions of C_2 radicals reacting with air. From plasma temperatures calculated based on LIBS, we predicted that the Si^+ should condense first and deposit as an Si layer below the carbon network, which arrives later.

From the above discussion, logically, for the sample cut in an Ar atmosphere, more carbon deposits should be found in the debris, because there is no oxygen present to help to remove it, so it re-condenses on the SiC, above the Si as a carbon network. Whereas in air, there is O_2 , which can remove C_2 radical as gaseous products before it can re-condense on the SiC surface as a carbon network. In fact, this carbon-based ablation material that forms in the Ar atmosphere is so thick that it masks the SiC substrate from the probe beam. However, in air the deposit is still thin enough to see some of the SiC substrate's Raman signal. This is consistent with size of the Si signal intensities measured under Ar (thick overlay of carbon) and in air (thin overlay of carbon).

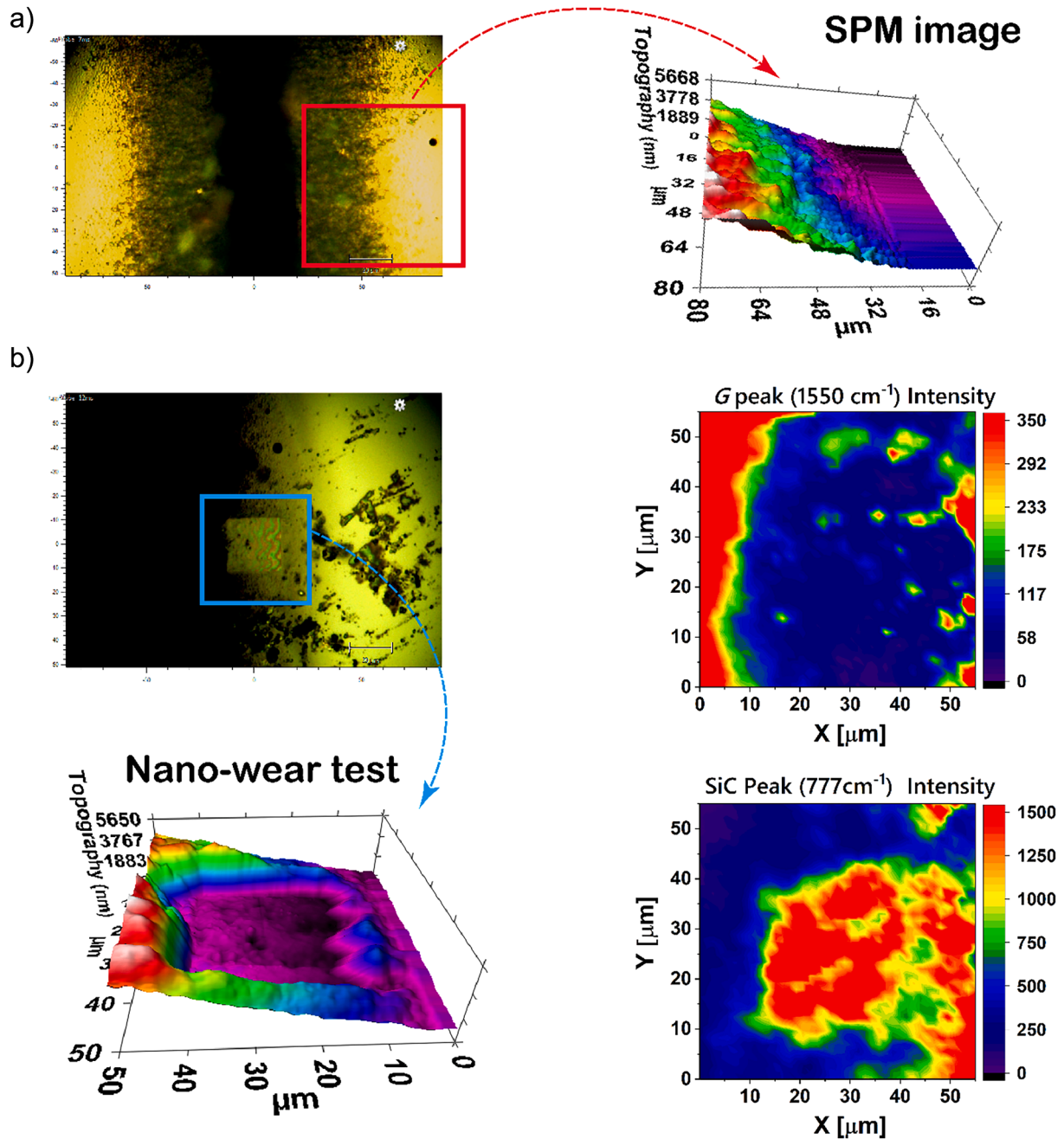


Fig. 7. (a) Optical and SPM images over the ablation debris in vicinity of a groove cut with 10 W laser power in air and (b) SPM image and corresponding Raman maps based on E_2 (TO) of SiC and G bands intensity over nano-wear test area over the built-up material in vicinity of a groove cut with 10 W laser power in air.

3.7. SPM and nano-wear

Fig. 7(a) shows the optical and SPM images over a built-up material in vicinity of a groove cut in air with 10 W laser power. It can be seen that the height of built-up material can reach to several microns and their height increases closer to the groove center. Fig. 7(b) shows an SPM and an optical image over a nano-wear test carried out on built-up material in vicinity of a groove cut in air with 10 W laser power along with co-located Raman maps over the nano-wear test area, based on E_2 (TO) of SiC and G band intensity. The Raman maps clearly confirm that built-up materials are loosely attached to the surface of the SiC wafer and can be scraped off with mechanical force. In other words, the darker deposits either side of the groove, the apparent HAZ, are not actually an HAZ at all. They are just recast deposits that can be mechanically

removed. The peak positions of the underlying SiC are not shifted from their original positions, meaning that the laser processing did not cause residual stress as a wide actual HAZ.

4. Conclusions

In this study, a nanosecond pulsed laser was used to examine the effects of assist gases during laser machining of 4H-SiC substrates. The average output power, scanning speed, and number of scans all influenced the process, in terms of groove width and depth (the volume of ablated material). Optical microscopy showed that the width and depth of grooves increased with increasing laser power and scanning times, however, the width of grooves reduced with increasing scanning speed because of a shorter laser-SiC material interaction time. According to the

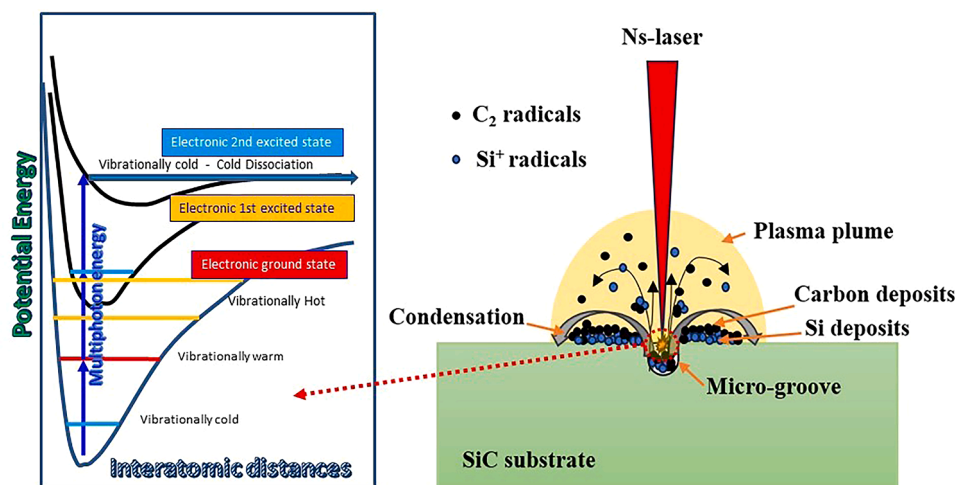


Fig. 8. Mechanism of SiC removal in plasma.

environment, narrow (Max. width $\sim 50\mu\text{m}$) and slightly deeper grooves are produced in Ar (Max. depth $\sim 170\mu\text{m}$), but much wider (Max. width $\sim 68\mu\text{m}$) and slightly shallower grooves are produced when machining in air (Max. depth $\sim 160\mu\text{m}$). The reason is a result of laser-induced decomposition reactions that occur during ablation in various environments.

The mechanism of ablation during laser-material interaction was studied using Raman spectroscopy to characterize the nature of the built-up material and using *In-situ* LIBS to record the events happening during laser ablation. Raman maps over the grooves show that the presence of O_2 in the environment leads to the gasification of laser-ablated carbon by the formation of CO_2 and CO molecules, leading to Si-rich a-C:Si deposits around the grooves. Co-located Raman maps over nano-wear sites confirm that the built-up material around the grooves is loosely attached to the SiC wafer surface and can be easily scraped off by mechanical force. Plasma temperature of Si^+ peaks and C_2 radicals inferred that C_2 existed in plasma for longer time and condensed over Si as it cools. The actual HAZ was not noticeable and internal stresses were not apparent from the Raman shifts of the underlying SiC substrate either side of the groove.

Clearly, laser machining in an air atmosphere produces better results in terms of the increased volume of material removed, whereas an argon atmosphere is favored for slightly deeper grooves (increase by $\sim 10\%$), or to achieve a high aspect ratio of grooves (~ 9.5). Comparison of the ablation debris deposited under different laser types will pave the way to get deeper insights into the mechanism of laser-material interactions during laser ablation. In the future, our research will be extended to try different assist gases (apart from Ar) to know their effect during laser ablation using nanosecond laser. From a practical industrial point of view, LIBS is a relatively cheap methodology that, based on this study, could be routinely used to monitor and quality control any laser cutting technique that produces a plasma.

Based on all the information provided in this comprehensive study, the full mechanism of laser material erosion (ablation) can be summarized as shown in Fig. 8, and results from multiphoton absorption of the nanosecond laser pulse and not thermalization of the laser energy into the SiC substrate.

CRediT authorship contribution statement

Dileep Karnam: Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Mehdi Rouhani:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jonathan Hogley:** Writing – review & editing, Writing – original draft, Methodology, Investigation,

Conceptualization. **Chia-Hung Hung:** Supervision, Methodology. **Yu-Lung Lo:** Supervision, Resources, Methodology, Funding acquisition. **Sai Bhavani Sravan Metla:** Software, Investigation, Formal analysis, Data curation. **Yeau-Ren Jeng:** Supervision, Resources, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Author statement

Please check the following as appropriate:

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

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Supplementary materials

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